Statistics & Probability

Section 1: Counting Principles & Probability

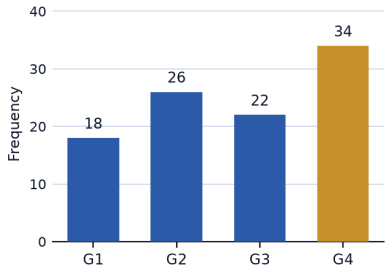
1 A password consists of 3 letters followed by 2 digits. If repetition of letters is NOT allowed, how many such passwords can be created?

a. $1,560,000$

b. $1,757,600$

c. $1,404,000$

d. $2,600,000$



2 Five people are to be seated in a row of five chairs. What is the probability that two specific people, Alice and Bob, sit next to each other?

a. $\frac{2}{5}$

b. $\frac{1}{5}$

c. $\frac{1}{2}$

d. $\frac{3}{5}$

3 What is the probability of obtaining exactly 3 heads in 5 tosses of a fair coin?

a. $\frac{5}{16}$

b. $\frac{1}{2}$

c. $\frac{3}{16}$

d. $\frac{5}{32}$

Section 2: Conditional Probability & Independence

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4 If $P(A) = 0.4$, $P(B \mid A) = 0.75$, and $P(B) = 0.5$, what is $P(A \mid B)$?

a. 0.60

b. 0.50

c. 0.30

d. 0.75

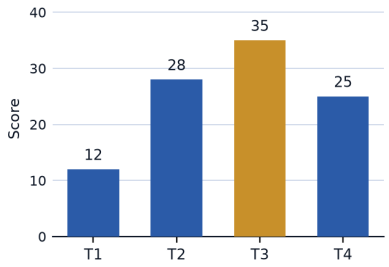
5 Two events A and B satisfy $P(A) = 0.3$ and $P(B) = 0.6$. If A and B are independent, what is $P(A \cup B)$?

a. 0.72

b. 0.90

c. 0.78

d. 0.18



6 In a certain school, 60% of students take French, 40% take Spanish, and 20% take both. What is the probability that a student takes French given that they take Spanish?

a. 0.50

b. 0.33

c. 0.60

d. 0.25

Section 3: Normal Distributions & Standard Scores

7 In a standard normal distribution $Z \sim N(0, 1)$, what is $P(-1.96 \leq Z \leq 1.96)$?

a. 0.95

b. 0.68

c. 0.997

d. 0.90

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8 A dataset has mean $\mu = 100$ and variance $\sigma^2 = 25$. What score corresponds to a z-score of 1.8?

a. 109

b. 105

c. 118

d. 145

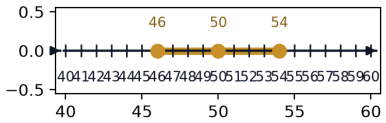
9 If 16% of scores on a normal test are above 84 and the mean is 76, what is the standard deviation?

a. 8

b. 4

c. 12

d. 16



Section 4: Regression & Correlation

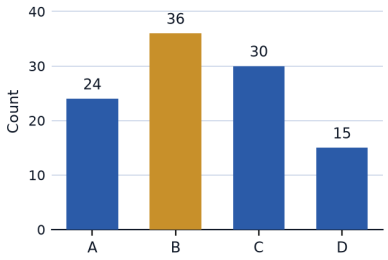
10 If the correlation coefficient between two variables is $r = -0.85$, what does this indicate?

a. A strong negative linear association

b. A weak negative linear association

c. A strong positive linear association

d. No linear association



11 In a least-squares linear regression line $\hat{y} = 2.5x + 14$, what is the predicted change in y for each unit increase in x?

a. $+2.5$

b. $+14$

c. $+5.6$

d. Cannot be determined

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12 An observed data point has coordinates $(4, 25)$. If the regression model predicts $\hat{y} = 23$ for $x = 4$, what is the residual?

a. $+2$

b. -2

c. $+48$

d. $+0.08$

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